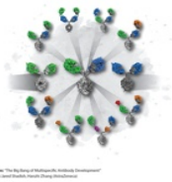
mAbs



ISSN: 1942-0862 (Print) 1942-0870 (Online) Journal homepage: www.tandfonline.com/journals/kmab20

Trispecific SEED antibodies engineered for neutrophil-mediated cell killing

Veronica Natale, Gergely Heves, Katharina Stadlbauer, Florian Rüker, Vanessa Siegmund, Lukas Pekar, Stefan Zielonka, Lars Toleikis, Stefan Becker & Gordana Wozniak-Knopp

To cite this article: Veronica Natale, Gergely Heves, Katharina Stadlbauer, Florian Rüker, Vanessa Siegmund, Lukas Pekar, Stefan Zielonka, Lars Toleikis, Stefan Becker & Gordana Wozniak-Knopp (2025) Trispecific SEED antibodies engineered for neutrophil-mediated cell killing, mAbs, 17:1, 2532851, DOI: 10.1080/19420862.2025.2532851

To link to this article: https://doi.org/10.1080/19420862.2025.2532851

© 2025 The Author(s). Published with license by Taylor & Francis Group, LLC.

View supplementary material

Published online: 15 Jul 2025.

Submit your article to this journal

Article views: 3333

View related articles

View Crossmark data

Citing articles: 4 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=kmab20

MABS
2025, VOL. 17, NO. 1, 2532851
https://doi.org/10.1080/19420862.2025.2532851

Taylor & Francis
Taylor & Francis Group

REPORT

OPEN ACCESS Check for updates

Trispecific SEED antibodies engineered for neutrophil-mediated cell killing

Veronica Natale[a], Gergely Heves[a*], Katharina Stadlbauer[a], Florian Rüker[a], Vanessa Siegmund[b], Lukas Pekar[b], Stefan Zielonka[b], Lars Toleikis[b#], Stefan Becker[b], and Gordana Wozniak-Knopp[a]

[a]Christian Doppler Laboratory for Innovative Immunotherapeutics, Institute of Molecular Biotechnology, Department of Biotechnology, University of Natural Resources and Life Sciences (BOKU), Vienna, Austria; [b]NBE Technologies, Global Research and Development, Merck Healthcare KGaA, Darmstadt, Germany

ABSTRACT

Immunoglobulin (Ig) A has attracted interest as a proposed therapeutic agent due to its ability to engage cell groups differently compared to an IgG scaffold and elicit tumor eradication. Further, its multimeric forms enable increased flexibility in the design of available paratopes. The latter is particularly advantageous for bi- and multispecific antibody formats, which are unparalleled in their enhanced selectivity and unique biological functions. We engineered bispecific heterodimeric IgA-based antibodies using the strand-exchanged engineered domain (SEED) technology, which relies on intertwined segments of IgA and IgG in the C_H3 domain, and applied mutagenesis to introduce two additional binding sites to enable the interaction of IgA-Fc with the myeloid cell-activating receptor CD89 (FcαR). These antibodies exhibited good biophysical properties and thermostability similar to the parental SEED molecule. Binding capacity to both antigens recognized by variable domains, epidermal growth factor receptor (EGFR) and receptor tyrosine kinase like orphan receptor 1 (ROR1), was not impaired, and in contrast to the original SEED-IgA, trispecific mutants could bind to CD89-expressing cells, mediate tumor cell-effector cell clustering, and induce neutrophil-mediated specific lysis of tumor cells. Trispecific design was applicable to both SEED-IgA1 and -IgA2 scaffolds. Interestingly, HEK-expressed mutants featured a CH2-linked N-glycan pattern more similar to wild-type IgA, with reduced core fucosylation in comparison with IgA-SEED. Collectively, the presented format combines the mobilization of CD89-positive effector cells with the flexibility of incorporating antigen specificities of choice into the variable domains, and thus is a promising basis for biochemically stable multispecific IgA with high therapeutic potential.

ARTICLE HISTORY
Received 22 April 2025
Revised 7 July 2025
Accepted 8 July 2025

KEYWORDS
ADCC; bispecific antibody; EGFR; Fc alpha RI; immunoglobulin a; neutrophils; ROR1; strand-exchanged engineered domain (SEED)

Introduction

Antibody-based therapies that engage in cell-mediated attack of tumor cells have proven to be one of the most efficient strategies in recent decades for combating cancer disease. Initially, these methods have relied on activation of natural killer (NK) cells via FcγRIII binding by IgG1-based therapeutics,[1,2] but the advent of bispecific antibodies has propelled potent T-cell activation as a mode of action into several clinical settings. Consequently, the majority of bispecific antibodies approved for human use today display engagement of T cells and simultaneous binding of a tumor-associated antigen as their main mode of action.[3] Nevertheless, these therapies are limited by the ability of the patient to respond either because of low numbers of T cells in the tumor microenvironment or T-cell exhaustion.[4,5] At the same time, severe side effects have been reported in many cases, including cytokine release syndrome, which has symptoms ranging from shivers and mild diarrhea to systemic inflammation and death, and diverse neurological symptoms.[6,7] Neutrophils have long been recognized as a cell group that can be activated by bispecific antibodies to attack cancer cells.[8–10] In contrast to T cells, they appear in high numbers and are present in the bloodstream as well as in

CONTACT Gordana Wozniak-Knopp gordana.wozniak@boku.ac.at Christian Doppler Laboratory for Innovative Immunotherapeutics, Institute of Molecular Biotechnology, Department of Biotechnology, University of Natural Resources and Life Sciences (BOKU), Muthgasse 18, Vienna 1190, Austria

*Current affiliation: GH: Boehringer Ingelheim RCV GmbH & Co KG,Doktor-Boehringer-Gasse 5–11, 1120 Wien.
#LT: Numab Therapeutics AG, Bachtobelstrasse 5, CH-8810 Horgen, Switzerland.

Supplemental data for this article can be accessed online at https://doi.org/10.1080/19420862.2025.2532851

© 2025 The Author(s). Published with license by Taylor & Francis Group, LLC.
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (http://creativecommons.org/licenses/by-nc/4.0/), which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

the tissues.[11] As they are a part of the innate immune system, their concentration and activity are less affected by tumor evasion mechanisms as discovered for T cells.[12] Most of the described neutrophil-activating bispecific antibodies mediate the formation of an immunological synapse between the tumor cell and neutrophils via the interaction with the receptor FcαR (CD89), expressed on their surface, and can incite potent tumoricidal activity via different mechanisms, including degranulation, neutrophil extracellular trap release, apoptosis, phagocytosis, and trogoptosis.[11,13] Importantly, there are no deficiencies described in the expression of CD89, in strong contrast with Fcγ receptors.[14] The extreme potency of the killing activity has been assigned to possible concurrent activation of neutrophils via CD89 and activating Fcγ receptors: combinations of target-specific antibodies of IgA and IgG class mediated more potent killing by polymorphonuclear cells (PMNs),[15] and IgG-type antibodies binding to tumor antigen and CD89 could engage NK cells, macrophages, as well as neutrophils, in the killing of cancer cells.[16] Tandem constructs of such type could cause more potent PMN-mediated killing than either of the parental IgG1 or IgA2 building blocks, used for construction, and exhibited good pharmacokinetic properties in mice.[17] In addition to fusions, "cross-isotype" IgG/IgA antibodies were constructed that could react with two activating Fcγ receptors as well as engaged FcαR and could incite more potent antibody-dependent cellular cytotoxicity (ADCC) and antibody-dependent cellular phagocytosis (ADCP) than the antibodies of IgG class.[18] Generally, it is believed that for potent cellular activation via CD89, cross-linking via bivalent interaction of the receptor and the IgA-Fc is required.[19,20]

In the past three decades, bispecific antibodies have not only emerged as effector-cell groups engaging molecules but also demonstrated superior selectivity[21] and the ability to overcome redundant signaling pathways,[22] and thus enabled more efficient intervention than antibodies of single specificity by addressing the same targets. As a result, there are more than 130 different formats available,[23] and one of the most diverse classes are asymmetrical heterodimeric bispecific antibodies, where targeting of two different antigens commonly relies on the two antibody Fab arms and the heterodimerization property of the Ig-C_H3 domains is steered toward pairing of complementary Fc-chains. This has been successfully achieved, for example, in the strand-exchanged-engineered domain (SEED) format, where the C_H3 domains comprise two heterologous chains named AG and GA, which are composed of interchanging amino acid stretches of IgG and IgA.[24] Such bispecific antibodies have been efficiently used to target cancer cells with enhanced selectivity,[25] shown enhanced activity resulting from biparatopic antigen binding,[26,27] and were able to activate T cells and NK cells upon the contact with tumor cells.[28] Typically, the paratopes within the SEED antibody are composed of a Fab fragment and a single-chain Fv (scFv) fragment, to avoid possible promiscuous pairing of the light chain, but this issue has been addressed either by exploiting a common light chain[29] or other technologies for preferential cognate heavy and light-chain pairing.[28,30]

Until now, SEED heterodimers have been expressed incorporating only IgG-C_H1 and C_H2 domains including the hinge region, but the attractive option of extending the spectrum of their mechanisms of action prompted us to explore their potential also in IgA context. Moreover, highly thermostable bispecific heterodimeric antibodies based on IgA-scaffold have been used recently to mediate close target-effector cell interaction.[31] We were further interested if the SEED-type heterodimeric C_H3 domains can be expressed as a part of the IgA scaffold and if such molecules can be further engineered to incite neutrophil-mediated effector functions.

Results

Molecular design

In the initial constructs, anti-epidermal growth factor receptor (EGFR)-Fab 225[32] and anti-receptor tyrosine kinase-like orphan receptor 1 (ROR1) scFv H082[33] were expressed fused with IgAC_H1, hinge, and C_H2 domain, and SEED GA- and AG-domain, respectively (sequences of all constructs in Supplemental Table S1). EGFR is a validated tumor target, addressed by several blockbuster therapeutic antibodies such as cetuximab and panitumumab.[34] The expression of ROR1 correlates positively with the progression of several blood[35] and solid malignancies,[36] and potentiates EGF-induced EGFR-signaling.[37] Although the heterodimeric C_H3 domains contain contigs corresponding to the IgA sequence, the residues discovered to be important for CD89 interaction[38] are modified.

The prominent interaction contig P93_L94_A95_F96 is replaced for HNHY in both SEED chains[24] with the purpose of retaining the neonatal Fc receptor (FcRn) binding[39] (numbering of the C_H3 domains are as proposed by Davis et al.[24] and in Figure 1a). Further, in IgA complexed with CD89, the C_H3-domain residues E90, L94, F96, M86, E47, S45, and R40 have been shown to play an important role for the interaction.[42] We initially targeted the residues whose mutagenesis could lead to an improved interaction of SEED C_H3-domains with CD89, using homology modeling in an

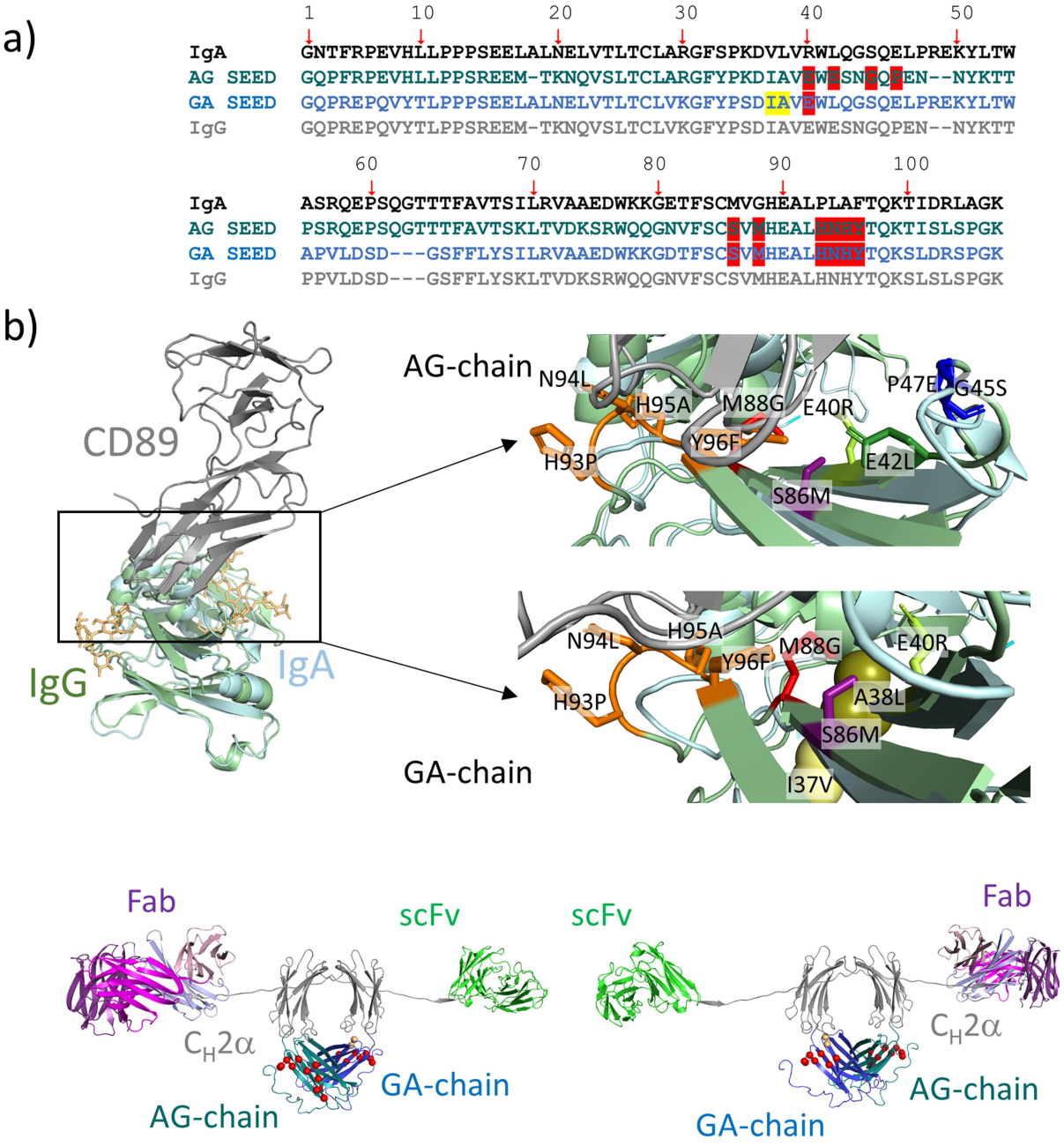


Figure 1. Molecular features of bispecific SEED-based antibodies with effector functions. (a) Sequence alignments of C_H3 domains of IgA1, IgG, SEED-AG chain and SEED-GA chain. Amino acid residues that were subject to modification to mediate CD89 binding are highlighted in red and those altered to influence the thermostability in yellow. (b) Overlay of IgA1-Fc (cyan) with bound CD89 (gray) (sugar residues in pale orange) (PDB:1OW0)[38] and IgG-Fc (green) (PDB:1OQO) (top left), with altered residues in the C_H3 domain highlighted (top right), and cartoon diagram of mutated SEED construct (bottom) with amino acids modified to mediate CD89 binding in red and those to influence thermostability in yellow (front and back view) (PDB: 1IGA).[40] the Figure was prepared with PyMOL molecular graphics system, version 2.4.0 Schrödinger, LLC, using PDB: 1IGA, and scFv was modeled with SWISS_MODEL.[41]

overlay of the structures PDB:1OW0[38] and PDB:1OQO and searched for amino acids interacting with CD89 within a distance of 4 Å to reversely mutate them.

In the SEED GA-chain, the stretch of residues H93_N94_H95_Y96 has been altered to P93_L94_A95_F96 (Figure 1b). S86 has been exchanged for M as the mutagenesis of this residue to A has been reported to render the IgA about 100-fold less active in binding to CD89. Also, M88 has been altered to G, as it could otherwise significantly impact the mobility of L257 and L258 in the C_H2 domain (numbering for the IgA sequence is according to the myeloma IgA1 protein (Bur) scheme[43]). E40 has been mutated to R to counter charge reversal and restore the stabilizing contacts with a part of the N263-linked glycan that can be expected judging from the crystal structure.

In the SEED AG-chain, the same changes have been introduced. Additionally, G45 was changed to S and P47 to E as both are positioned within 4 Å from L54 in CD89. Specifically, E47 forms a polar contact with a galactose residue on the N263-linked glycan. At this point, an antibody containing all these mutations (MUT2210) was produced, and indeed, a weak interaction with CD89 could be detected in a biolayer interferometry experiment (Supplemental Figure S1).

Moreover, the AG-chain was modified in that the E42 was changed to L, as this residue is also within 4 Å from the residues L54 and K55 in the CD89 (residue numbering of CD89 as in 1OW0).[38]

Finally, the residues I37 and A38 were targeted for mutagenesis in both chains, as it has been shown previously that changes at these positions may influence the stability of the C_H2 domain in the IgG context.[44] While this mutagenesis in the GA-chain caused an increase in affinity for CD89, it was not beneficial for the CD89 binding when included in the CD89-binding variant of the AG-chain (Figure S1).

To enable efficient expression and avoid multimerization, C-terminal residues C and Y were removed from the coding sequence in all constructs.[45,46]

In the IgA2 scaffold, the light chains are not covalently linked with the heavy chains, and therefore the interaction of light- and heavy-chain interaction was stabilized by introducing the P221R mutation for all expressed mutants.[47]

Expression and purification of SEED-IgA variants

225-IgA1 and SEED-IgA1 variants were expressed at a yield of 60 mg/L after Protein L purification and contained about 72% monomeric species (Figure 2a). This percentage was notably higher for both, IgA1 and SEED-IgA1, than when those were expressed with a full-length C-terminus harboring end C and Y residues (41% and 38%). The SEED-IgA1 constructs were of high thermostability, showing a first thermal transition at about 65°C, associated with the denaturation of modified C_H3 domains, and a second at about 75°C, corresponding to overlapping denaturation of Fab and C_H2 domains (Figure 2b). Importantly, the mutagenesis required for the introduction of a CD89-binding site did not negatively impact thermostability properties (Figure S1). Consequently, the SEED variant with the highest affinity for CD89 named MUT2724 was chosen for further characterization. In SDS-PAGE analysis, all three proteins exhibited a single main band (Figure 2c). SEED-IgA1 additionally displayed a band at about 75 kDa, which probably corresponds to a single GA-heavy chain and a light chain, and another one at 60 kDa, probably corresponding to a single AG-heavy chain, and interestingly these were less present in MUT2724-IgA1, indicating its higher chemical stability.

Glycan analysis

We labeled the glycosylation sites for both IgA1 and IgA2 with I-IV according to their sequence homology (Figure 3a). In all IgA1-based antibodies, the glycosite containing N263 (Site II) could be analyzed (Figure 3b). Interestingly, the unmodified SEED variants contained a markedly higher proportion of fucosylated glycans (71.24%) than wild-type 225-IgA1 (1.81%) and MUT2724 (6.86%). Glycan composition at the glycosite at N459 (Site IV) was more similar between the SEED and MUT2724, for both GA and AG-chain. While this site was not used in about 50% of the AG-chain and in about 40% in the corresponding fragment of the wild-type IgA1, only 15–20% of the GA-chain pertaining fragments were not glycosylated.

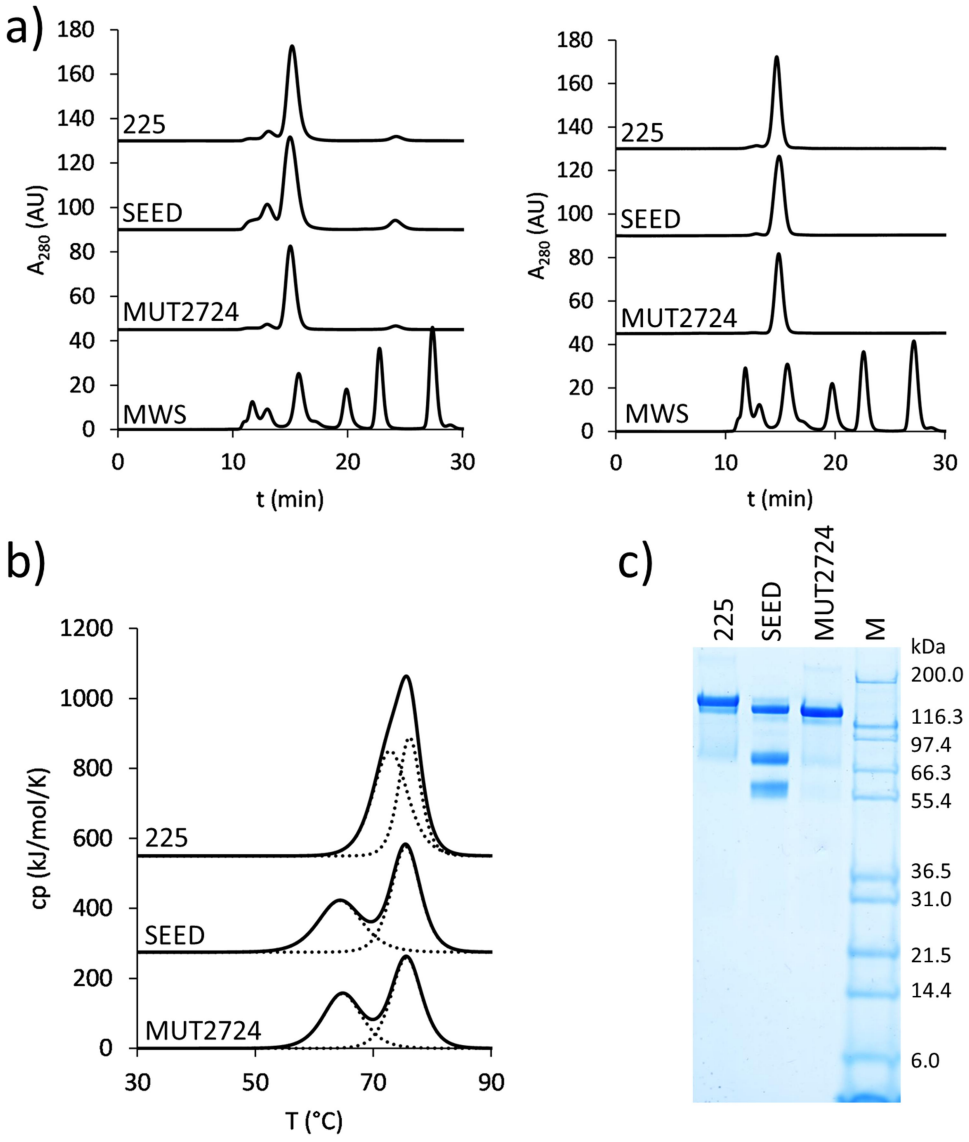


Figure 2. Biophysical characterization of IgA- and SEED-based constructs. (a) Analysis with size exclusion chromatography in native conditions before (left) and after (right) gel filtration (MWS: molecular weight standard with peaks of 670, 158, 44, 17 and 1.3 kDa). (b) Thermostability analysis using differential scanning calorimetry. (c) SDS-PAGE analysis (M: mark 12 unstained marker).

Analysis of antigen- and cell-binding properties

In contrast to SEED-IgA1, MUT2724-IgA1 could bind to recombinant CD89 with 57 nM affinity, in comparison with 27 nM determined for 225-IgA1 (Figure 4a). Importantly, the novel mutant could bind simultaneously to both tumor-associated antigens (Figure 4a). The mutagenesis applied to promote CD89 binding did not impair the binding properties to any of the cognate antigens of the bispecific SEED antibody and antigen-positive cell binding remained the same, with an EC_{50} of about 2 nM for both molecules (Figure 4b). Internalization of MUT2724-IgA1 proceeded faster than that of homodimeric 225-IgA1, to 42% vs. 32% after 4-h-incubation and 68% vs. 57% after overnight incubation, while SEED-IgA1 was 31% after 4 h and 71% after overnight incubation (Figure 4c). For the analysis of binding to cell-expressed CD89 antigen, induced HL-60 cells were used. The number of the CD89 copies was determined to be 8000 per cell (Supplemental Figure S2), which was similar to the number measured for neutrophils (Supplemental Figure S3) and also corresponds well to the value reported for neutrophils in the literature.[50] While SEED-IgA1 was not reactive with the induced cells, MUT2724 bound to the cell surface as well as the positive control 225-IgA1 (Figure 4d). The interaction was dependent on the CD89 expression, as only background reactivity

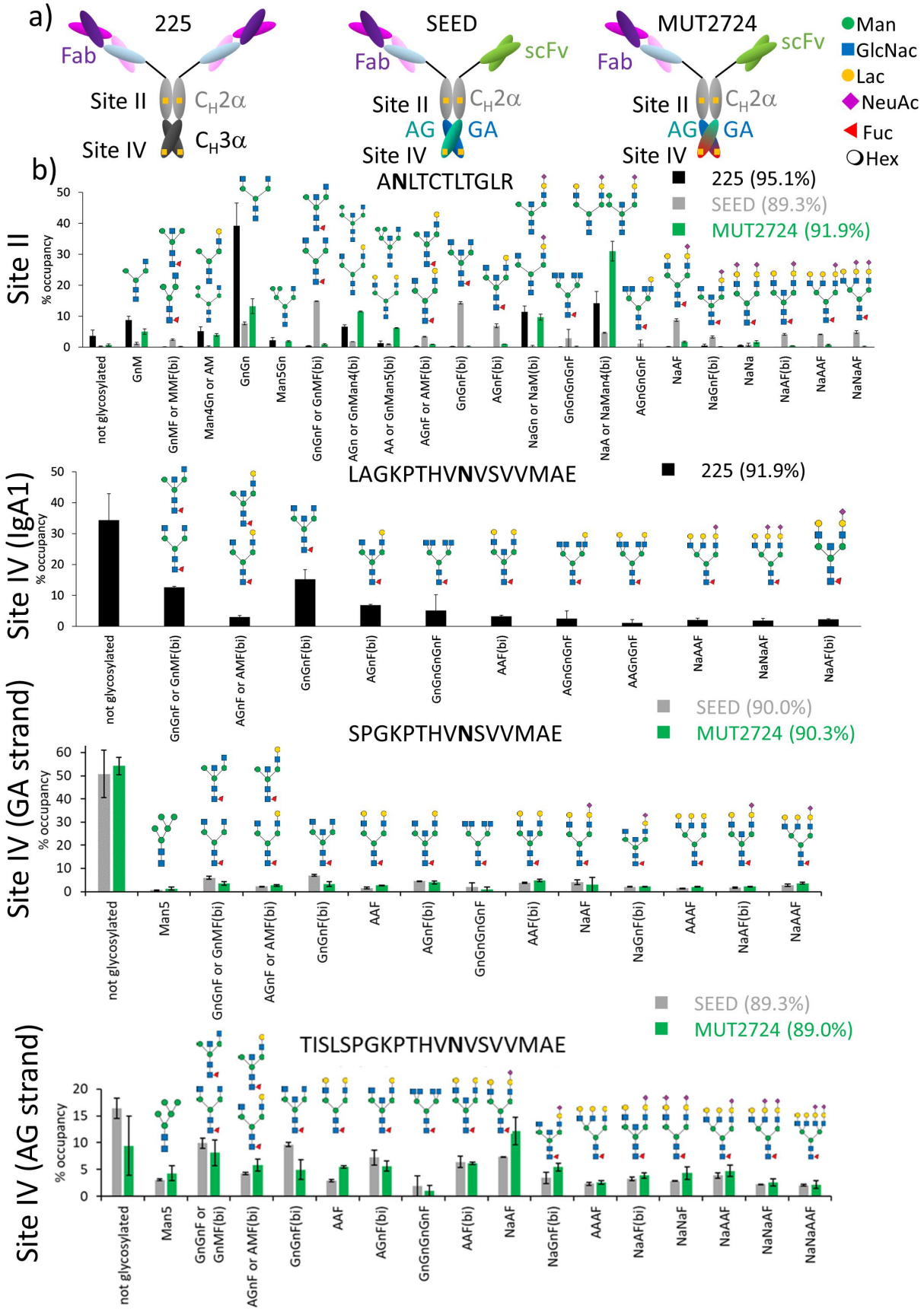


Figure 3. Analysis of glycans in IgA1, SEED-IgA1 and mutated SEED-IgA1 constructs. (a) Schemes of the individual constructs showing the analyzed glycosites (yellow square) and legend of symbols used for presentation of glycan patterns.[48] (b) frequency of glycans found at individual glycosites (percentage in brackets indicates the coverage achieved after all glycoforms with frequency above 2% were included). The Figure was prepared using GlycoGlyph.[49].

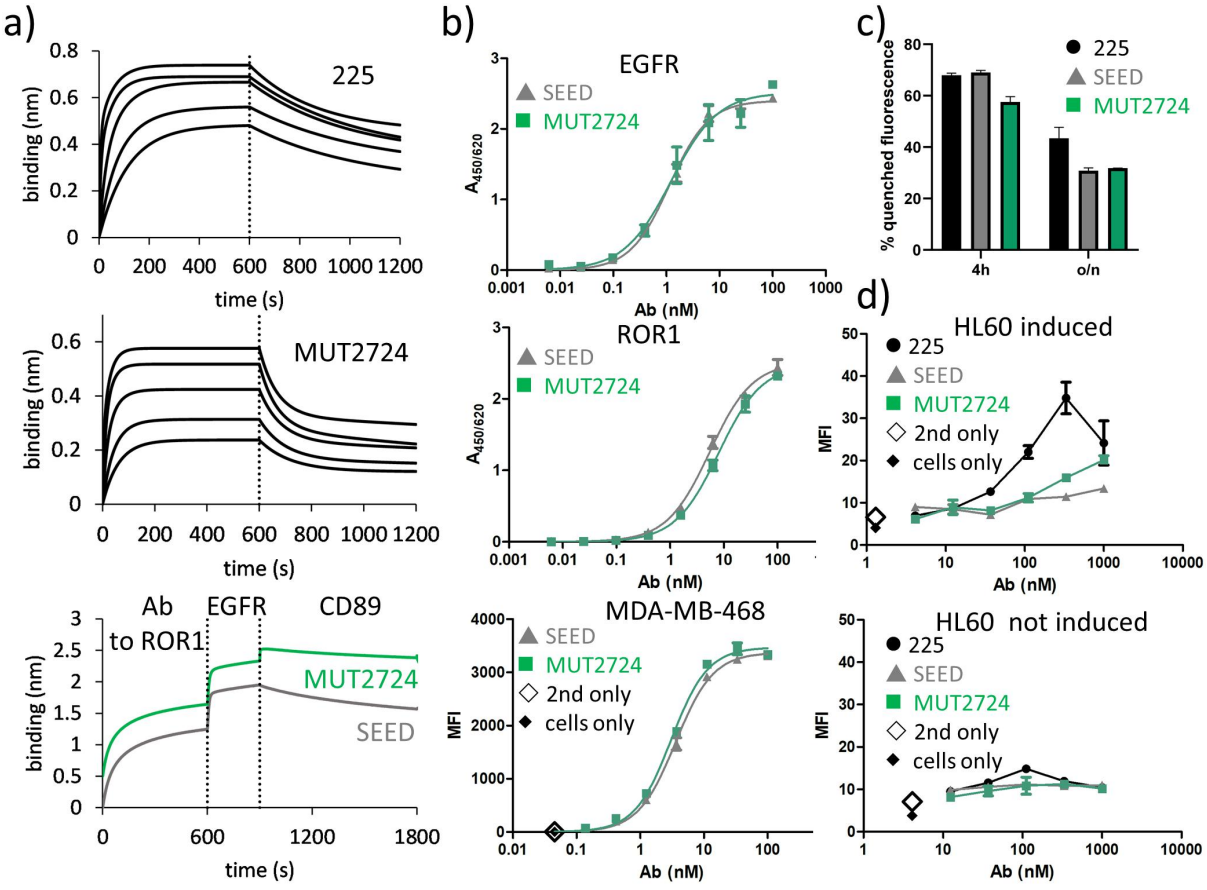


Figure 4. Antigen-binding properties of IgA1, SEED-IgA1 and mutated SEED-IgA1. (a) Biolayer interferometry experiment showing binding of 225-IgA1 (top) and mutated SEED-IgA1 (center) to immobilized CD89), and difference in binding to CD89 between the mutated and wild-type SEED-IgA1 after the reaction with ROR1 and EGFR (ab: antibody). (b) ELISA assay showing binding of SEED-IgA1 and mutated SEED-IgA1 to EGFR (top) and ROR1 (center) and FACS assay of binding to antigen-positive cell surface. (c) Internalization of the constructs into MDA-MB-468 cells monitored with percentage of quenched surface fluorescence. (d) Binding of the constructs to the surface of cell line HL-60 when induced for expression of CD89 and before induction (2nd only: secondary reagent only).

was measured with non-induced HL-60 cells. The hook-effect appearing with the concentrations of 225-IgA above 300 nM probably results from the depletion of binding sites that allow bivalent binding, and the start of prevailing monovalent interaction with the homodimeric IgA, which is two orders of magnitude weaker.[38,51,52]

Cell–cell interaction assay and neutrophil activation

The ability of MUT2724-IgA1 to mediate strong intercellular contacts was examined in a cell–cell interaction assay. Upon co-incubation of target and effector cell types, where induced HL-60 cells were used as effector cell type and MDA-MB-468 as target cell type with 20 nM antibody, about 30% of the measured events were complexes of both cell types, in comparison with 35% observed with 225-IgA1 (Figure 5a). The presence of SEED-IgA1 did not change the percentage of associated cells from that formed without any antibody (15% vs. 13%). For further evidence of the specificity of this assay, we have produced trastuzumab (TRA)-IgA1. This antibody mediated association of 16% of the cells, which was not above background for the Her2-negative cell line used. None of the antibodies had any effect when HEK293-6E cells were used as

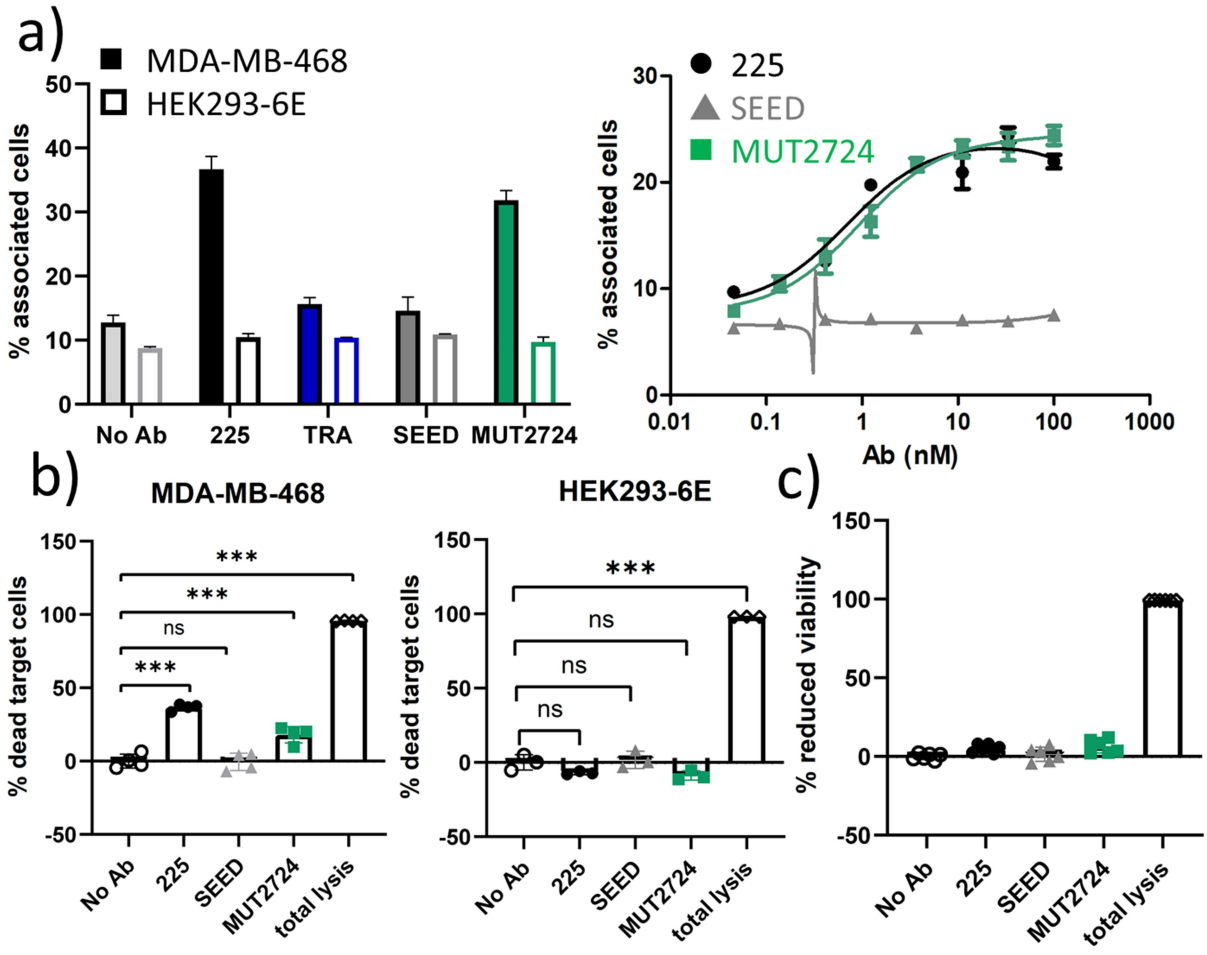


Figure 5. Biological activity of IgA1-based constructs. (a) Cell–cell interaction assay evaluated for the percentage complexed tumor target cells and induced HL-60 cells with or without addition of the antibodies (left) and the dose-dependency of antibody concentration (right). (b) Results of antibody-dependent cytotoxicity assay using neutrophils as effector cells, with antigen-positive MDA-MB-468 cells and control HEK293-6E cells. (c) Direct effect of antibodies on MDA-MB-468 cells (no effector cells). Significance was determined with one-way ANOVA analysis with GraphPad prism 8.0.2, ns: not significant, $p > 0.05$; *: $0.05 > P > 0.01$; **: $0.01 > P > 0.001$; ***: $p < 0.001$.

a target cell line. The ability to mediate cell–cell association was dose dependent in the concentration range from 0.1 to 10 nM.

Finally, we were interested if the novel mutant can indeed induce neutrophil-mediated killing. Neutrophils were isolated at a high purity (Supplemental Figure S3). In contrast to SEED-IgA1, which was inactive in this assay, MUT2724-IgA1 killed 21% of the target cells, while 225-IgA1 was more potent, with 37% killed cells at 20 nM concentration (Figure 5b). There was no effect when antigen-negative HEK293-6E cells were used as a target cell line. The mere addition of the proteins to the culture medium had no effect on the MDA-MB-468 target cell line (Figure 5c).

Neutrophil-engaging SEED antibody in IgA2 format

Compared with IgA1, the IgA2 format is believed to be better suited for potential clinical development due to certain advantageous features, such as shorter hinge region, which hinders digestion by microbial proteases,[53] and its lack of several O-linked glycosylation motifs present in IgA1.[54] We introduced the mutations required for CD89 engagement into a SEED-IgA2m(1) scaffold with the same antigen specificities as described above (Supplemental Figure S4) and examined the features of the CD89-binding SEED-based mutant (MUT2724-IgA2) also in this molecular context. The antibodies 225-IgA2, SEED-IgA2, and MUT2724-IgA2 could be purified at 60 mg/L supernatant with Protein L-based affinity chromatography,

and monomer amounted to 43% for SEED and 84.5% for Mut2724 (Figure 6a). After the gel filtration, 17 mg of monomeric antibody per liter supernatant were obtained. The thermostability of the antibodies was high, with deconvolutions measured at 76.3°C and 80°C for 225-IgA2, 64.9°C and 78.8°C for SEED, and 65.8°C and 78.6°C for MUT2724 (Figure 6a). The produced proteins appeared as one major band in the SDS-PAGE analysis, and again the minor bands of lower molecular weight were less pronounced for MUT2724 than for SEED. As in the IgA1 background, the fucosylation of core N263-glycan (Site II) in SEED was more common than for 225-IgA2 and MUT2724-IgA2 (14.9% vs. 0.32% and 1.04%, respectively) (Supplemental Figure S5). Site III (including N337) was occupied with similar structures in all three mutants. As in the IgA1-based species, Site IV was not used in about 50% of the AG-chain, as seen also in the corresponding fragment of the wild-type IgA2, but in contrast only 15–20% of the GA-chain pertaining fragments were not glycosylated.

In an IgA2 scaffold, bispecific MUT2724 could bind to CD89 with an affinity of 48 nM, whereas 31 nM was measured for 225-IgA2 construct (Figure 6b).

Target antigens EGFR and ROR1 were bound with an EC_{50} of 1.1 nM and 10.0 nM by SEED vs. 1.1 nM and 7.2 nM by MUT2724 and binding to MDA-MB-468 cells proceeded with an EC_{50} of 4.4 nM for SEED and 2.9 nM for MUT2724 (Figure 6c); interestingly, here the maximal level of binding was lower for MUT2724 than for SEED, which could be attributed to the lower mobility of the Fab arms and the scFv moiety as observed in IgA1 format. In contrast to SEED, MUT2724 could react with the induced HL-60 cell line (Figure 6d) and mediate strong target-effector cell association causing about 23% of the cells to cluster, similarly to 225-IgA2 (Figure 6e), while there was no effect on the control HEK293-6E cell line in this experiment. When the target MDA-MB-468 cells were incubated overnight together with antibodies and neutrophils, MUT2724-IgA2 could kill about 13% of the cells, 225-IgA2 killed 27% and the SEED-IgA2 was not active at all (Figure 6f). None of the mutants had an effect on HEK293-6E cells (Figure 6f) used as a control cell line, and there was also no direct toxicity (Figure 6g).

Discussion

We constructed bispecific IgA antibodies where heterodimerization is achieved via the SEED architecture of C_H3 domains, which was modified to enable the interaction with CD89. The molecules were well expressed in HEK293-6E cells in standard laboratory conditions and could be purified using Protein L affinity chromatography followed by preparative size-exclusion chromatography (SEC). While the original SEED-based constructs did not interact with CD89, mutagenesis of nine residues in the SEED GA-chain and ten residues in the AG-chain appreciably delivered molecules that could bind to this important ligand. Resulting antibodies could also mediate the formation of target and effector cell clusters and induce neutrophil-mediated specific target cell killing. As such, the proposed format therefore represents an important addition to the spectrum of IgA-based anti-cancer reagents, whose potential was only recognized in the last decades.[55] Only recently, it has been reported that the expression of CD89 is increased in nearly all myeloid cell subsets in tumor-bearing mice transgenic for human CD89 and that this ligand is expressed on tumor-infiltrating myeloid cells.[56] Bispecific antibodies that engage CD89 and the tumor marker ALPPL2 and are inert toward Fc gamma receptors, efficiently inhibited growth of ALPPL2-positive tumors in human CD89-transgenic mice by enhancing macrophage-mediated phagocytosis of tumor cells. Moreover, these antibodies not only shifted tumor-associated macrophages toward a pro-inflammatory M1 phenotype but also increased the infiltration of CD8-positive cells,[57] which underlines the promise of CD89-reactive antibodies for cancer immunotherapy. The affinity of the novel constructs toward CD89 was in the same range as that determined for the IgA1 and IgA2 control constructs with cetuximab (225) variable domains, and these in turn are reported not to differ from unmodified IgA molecules without any engineered mutations.[20,46,47] This is important because otherwise the presence of wild-type IgA could impair the killing effect.

With the mutagenesis strategy used here, the model anti-EGFR/anti-ROR1 IgA-SEED antibody did not lose the ability to recognize its original antigens. As the CD89 interaction is based on the recognition motifs in the Fc regions, this method can be considered applicable to incorporate antigen specificities of choice. For example, an element blocking immunomodulatory CD47-SIRPα signaling could prove valuable to further augment the activation of myeloid cells to kill target cancer cells.[58] Apart from cancer targeting, IgA

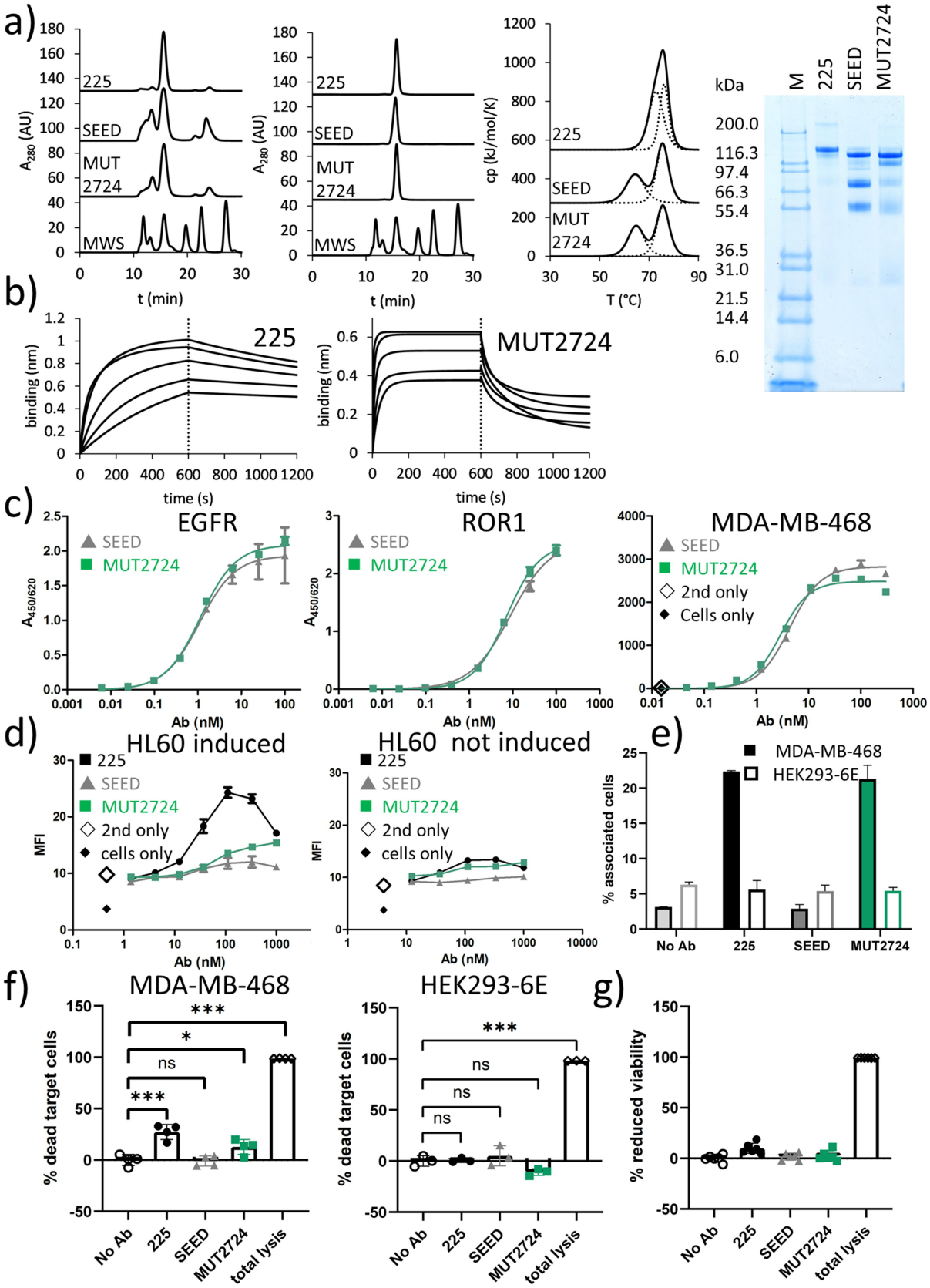


Figure 6. Characterization of antibodies in IgA2 scaffold. (a) Analysis with size exclusion chromatography in native conditions before and after gel filtration (left) (MWS: molecular weight standard), thermostability determined with differential scanning calorimetry (center), and SDS-PAGE analysis (M: mark 12 unstained marker). (b) Biolayer interferometry experiment showing binding to immobilized CD89 by 225-IgA2 and mutated SEED-IgA2. (c) Binding of SEED- and mutated

antibodies have gained attention as therapeutically meaningful anti-microbial entities[59,60] and antiviral agents,[61] and here the use of bispecific antibodies could broaden the targeting spectrum by enhancing strain coverage.[62] It is also worth mentioning that, for both unmodified SEED-IgA and CD89-engaging MUT2724-IgA with our chosen specificities of variable domains, a higher level of internalization was determined than for 225-IgA for both IgA-subclasses. Apart from the lower affinity toward CD89, this could also have an adverse effect on the potency in the killing assays.

One important property that could contribute to the establishment of IgA-based therapeutics is the extension of plasma half-life, which is typically measured in days rather than weeks, as is characteristic for IgG1, 2- and 4-class antibodies owing to their recycling mediated by FcRn.[63] An engineered IgA2-based format (IgA 2.0), which in mice exhibits multiple times higher serum levels compared with unmodified IgA2, has been successfully designed.[46] While the IgG-SEED format retains its FcRn binding activity,[24] this cannot be expected for the CD89-engaging variant where the critical interaction residues H435 in both heterodimer chains have been altered as the interaction sites of CD89 and FcRn receptors with IgG-based molecules are largely overlapping. The pharmacokinetic properties of IgA format have been improved by exploiting the pH-dependent interaction of albumin with FcRn and designing a fusion with an albumin-binding domain, hence exploiting the pH-dependent interaction of albumin with FcRn,[64] or more recently with the whole human serum albumin, which has translated into 10- to 11-fold extended plasma half-life.[65] This could be the most straightforward strategy also for the newly designed molecules described here.

In the IgA2.0 format, the altered glycosylation pattern featuring a larger ratio of terminally sialylated N-glycans renders the antibody less likely to be captured and cleared by asialoglycoprotein receptor, which has an additional positive effect on the half-life *in vivo*. We also observed altered glycosylation of CD89-engaging SEED-IgA1 antibody, most prominently featuring less fucosylation of the N-terminal core than the SEED-IgA, especially at the glycosite containing N263, which is proximal to the CD89 binding site. This phenomenon is interesting because it reflects the activity of FUT8 enzyme, which is considered to depend on the accessibility and nature of the N-glycan and much less on the underlying peptide sequence.[66] As we have not introduced any changes in the C_H2 domain, it appears that solely its conformation, dependent on the status of SEED-C_H3 domain mutagenesis, influenced the level of FUT8-mediated fucosylation. Glycosylation at the N263 site itself is considered important for the interaction with CD89,[67] and moreover, aberrant N-linked glycosylation of IgA has been implicated in pathological conditions such as IgA nephropathy.[68]

While data on the immunogenicity of the IgA-based formats are currently limited, this is likely to change in the near future due to recent technical developments that facilitate their expression and manufacturing and support their faster translation in clinical settings.[46,69] Similar as in the parental bispecific SEED antibody, our proposed format contains only human sequences and differs within the intertwined IgA/IgG strands only in 19 amino acids, which is less than 10% of the C_H3 domain sequence.

In comparison with the existing trispecific CD89-addressing format Trisomab,[16] our antibody has CD89 binding and additionally two further antigen-binding sites on the Fab arms, which should allow targeting with better selectivity for the tumor cell. The interaction with CD89 is in our case one antibody to two CD89, which should enhance agonistic signaling.[20] The heterodimeric IgA format described by Heinkel *et al.* [31] is more similar to our construct. We think, however, that the unique design features of each one, such as glycosylation or Fab-Fc functional domain interplay, can trigger or limit a diverse set of Fc-mediated effector functions and therefore add to the opportunities for the design of antibodies with tailored functional profiles.[70]

SEED constructs to EGFR and ROR1 in ELISA and their reactivity with the MDA-MB-468 cell surface (2nd only: secondary reagent only). (d) Binding to cell surface of HL-60 cells after and before induction of CD89 expression (2nd only: secondary reagent only). (e) Cell–cell interaction assay evaluated for the percentage complexed tumor target cells and induced HL-60 cells with or without addition of the antibodies. (f) Activity of the antibodies in an ADCC assay using neutrophils as effector cells. (g) Direct effect of antibodies for MDA-MB-468 cells (no effector cells). Significance was determined with one-way ANOVA analysis with GraphPad prism 8.0.2, ns: not significant, $p > 0.05$; *: $0.05 > p > 0.01$; **: $0.01 > p > 0.001$; ***: $p < 0.001$.

In summary, we constructed a heterodimeric IgA-SEED-based antibody using rational mutagenesis, which can additionally incite neutrophil-mediated killing and displays a comparable stability to the parental SEED-IgA molecule. Due to the flexibility in the choice of antigens targeted with its Fab-arms, this scaffold can be considered a platform for the design of multispecific IgA-based therapeutic molecules.

Materials and methods

Expression and purification of IgA-SEED variants

Heavy and light chain constructs were cloned into pTT-based vectors (Canadian National Research Council (CNRC)). Mutations in the SEED domain were introduced by site-directed mutagenesis using the QuikChange Lightning Mutagenesis Kit (Agilent). IgA1, IgA2, and the corresponding IgA-SEED variants and derived mutants were expressed in HEK293-6E cells using a heavy-to-light chain ratio of 1:1 for homodimeric or 2:1:2 (AG-chain: GA-chain: light chain) for heterodimeric antibodies. Expression constructs have been introduced to the cells at a density of 1.5-2x10^6/mL using polyethyleneimine (PEI)-mediated transfection, with a mass ratio of DNA:PEI of 1:2. Cells were kept on orbital shaker at 125 rpm in hydrated atmosphere with 5% CO_2 at 37°C and cultured for 5 d with the addition of 0.5% TN-1 on day 2 after transfection. Supernatant was harvested with centrifugation at 2500 *g*, 15 min at 4°C, buffered with 0.02 M sodium phosphate/150 mM NaCl, pH 7.0, and cleared through a 0.45-µm-filter before the application to the Protein L column (Cytiva) equilibrated with the same buffer. After loading was completed, the bound protein was eluted with 0.1 M glycine, pH 3.0, and protein containing fractions were neutralized immediately with the addition of 19 µL 2 M Tris, pH 9.0, and dialyzed against at least 100-fold volume of phosphate-buffered saline (PBS), pH 7.4 at 4°C overnight. Monomeric IgA fraction was separated from oligomers and free light chain on a preparative gel-filtration column Superdex 200 Increase 10/300 GL (Cytiva) in PBS/200 mM NaCl, at 0.75 mL/min flow rate, and stored at −80°C. The run was monitored with Gel Filtration Standard (BioRad).

Biophysical characterization

Differential scanning calorimetry

About 5 µM protein solution, diluted in PBS, was heated in the Automated MicroCal PEAQ-DSC system (Malvern Panalytical) from 20°C to 100°C at a rate of 1°C/min, with repeated scans run at the same conditions to obtain the baselines for subtraction from the first scan. Data were fitted with Origin 7.0 for the DSC software, and the non-two-state transition mechanism was applied for scanning deconvolution and determination of midpoint temperatures of transitions (T_Ms).

Glycan analysis

The samples were digested in-solution. The proteins were S-alkylated with iodoacetamide and digested with Trypsin and GluC (Promega). The digested samples were loaded on a nanoEase C18 column (nanoEase M/Z HSS T3 Column, 100Å, 1.8 µm, 300 µm ×150 mm, Waters) using 0.1% formic acid as the aqueous solvent. A gradient from 1% B (B: 80% acetonitrile, 0.1% formic acid) to 40% B in 50 min was applied, followed by a 10 min gradient from 40% B to 95% B that facilitates elution of large peptides, at a flow rate of 6 µL/min. Detection was performed with an Orbitrap MS (Exploris 480, Thermo) equipped with the standard H-ESI source in positive ion and DDA mode (switching to MS/MS mode for eluting peaks). MS-scans were recorded (range: 350–1200 Da), and the 20 highest peaks were selected for fragmentation. Instrument calibration was performed using Pierce FlexMix Calibration Solution (Thermo Scientific). The possible glycopeptides were identified as sets of peaks consisting of the peptide moiety and the attached N-glycan. The theoretical masses of these glycopeptides were determined with a spreadsheet using the monoisotopic masses for amino acids and monosaccharides.

Manual glycopeptide searches were made using FreeStyle 1.8 (Thermo). For the quantification of the different glycoforms, the peak intensities of the deconvoluted spectra were compared. All the glycoforms present in over 2% were taken into the calculation.

Antigen-binding properties

ELISA

Biotinylated extracellular domains of ROR1 and Fc-tagged extracellular domains of EGFR (both Sino Biological) were coated to the wells of Streptavidin Immobilizer plate (NUNC) or Maxisorp plate (NUNC) at 5 µg/mL and 10 µg/mL in 100 µL PBS for 1 h at room temperature (RT). After washing 3 times with 200 µL PBS, wells were blocked with 200 µL 4% bovine serum albumin (BSA)-PBS for 1 h at RT. Antibodies were added in three-fold dilution series starting from 100 nM in 2% BSA-PBS and allowed to bind for 1 h at RT. After three washing steps with 200 µL PBS, their binding was detected for 30 min with 100 µL anti-human kappa-horseradish peroxidase conjugate (RRID: AB_258334, A-7164, Sigma-Aldrich), diluted 1:5000 in 2% BSA-PBS, and the color reaction was developed with 100 µL 3,3′,5,5′-3,3,5,5-tetramethylbenzidine (TMB) (Sigma-Aldrich) and stopped with the addition of 100 µL 30% H_2SO_4. Absorbance at 450/620 nm was recorded using a Tecan Spark reader.

Biolayer interferometry

Affinity to CD89. An Octet 96RED (ForteBio)-based assay was used for the determination of binding kinetics. The His-tagged extracellular domain (ECD) of CD89 (residue Q22-N227 (NP_001991.1) with C-terminal His6-Tag, expressed in HEK293-6E) was immobilized on HIS1K biosensors (Sartorius) and equilibrated in assay buffer (PBS supplemented with Kinetics Buffer) (ForteBio, Sartorius) at 10 µg/ml for 5 min. Antibodies diluted in an assay buffer in two-fold steps starting at 1 µM were allowed to bind to the immobilized receptor for 10 min at 25°C with shaking at 1000 rpm. Dissociation into the assay buffer was recorded for 10 min. Assay background was determined with a sensor immersed in an assay buffer only and background binding with antibodies binding to uncoated sensors. After background measurement subtraction, data were fitted with the 2:1 binding model (ForteBio Evaluation Software, version 11.0).

Simultaneous antigen binding. Streptavidin sensors were loaded with 10 µg/mL of biotinylated ROR1 (Sino Biological), and 100 nM antibody dilutions in the assay buffer were allowed to bind for 10 min. Sensors were then immersed into 10 µg/mL solution of EGFR-Fc (Sino Biological) in an assay buffer for 5 min, and finally into a solution of 100 µg/mL of His-tagged CD89-ECD in assay buffer for 15 minutes.

Binding to cell-bound antigens

Cultivation methods

MDA-MB-468 cells (ATCC HTB-132^{TM}) were cultivated in DMEM medium with 10% fetal calf serum (FCS) and penicillin/streptomycin and passaged twice a week. For detachment, the cells grown in a 75-cm^2 -T-flask were rinsed twice with 12 mL PBS, treated with 3 mL 0.5% Trypsin/EDTA solution (Sigma-Aldrich) for 5 min at 37°C, and pelleted at 300 *g*, 5 min at room temperature. They were then resuspended in 5 mL culture medium, counted, and either resuspended at the density indicated below in the culture medium for further propagation, labeling, internalization, and cell killing experiments or in 2% BSA-PBS for cell staining and cell–cell interaction assay. HL-60 cells (ATCC CCL-240^{TM}) were propagated in an RPMI medium with 10% FCS and penicillin/streptomycin and maintained at a density of 1-9x10^5 cells/mL.

Cell surface binding assays

***Expression of CD89 on HL-60 cells.*.** The expression of CD89 on the surface of HL-60 cells was induced with 1.3% dimethyl sulfoxide, and the peak expression was determined using staining with an anti-CD89 antibody (RRID: AB_447122, ab22520, Abcam) after 5 d. Briefly, 100,000 induced cells were blocked with 100 µL 2% BSA-PBS for 30 min on ice and then incubated with a 3-fold dilution series of anti-CD89, starting from 10 nM, in 2% BSA-PBS for 30 min on ice. After centrifugation at 300 *g* for 5 min at 4°C, binding was detected using anti-mouse-phycoerythrin antibody (RRID: AB_315010, 405307, Bio-Legend) and diluted at 1:500 in 2% BSA-PBS, for 30 min on ice. Finally, the cells were resuspended in 200 µL ice-cold PBS, and 5000 events were analyzed using a Guava easyCyte flow cytometer (Luminex). The number of copies of CD89 was determined using QIFIkit (Agilent) exactly according to the manufacturer's instructions, with the same primary anti-CD89 antibody.

The interaction of bispecific antibodies with target cells.. Surface target-expressing cells were diluted to 10^6 cells/mL in 2% BSA-PBS for blocking for 30 min on ice. 100-µL aliquots were distributed into the wells of a 96-U-well plate and pelleted at 300 *g* for 5 min at 4°C before the incubation with graded concentrations of antibodies, starting with 1000 nM for CD89 interaction on HL-60 cells and 100 nM for the determination of EC_{50} for binding to the surface of MDA-MB-468 cells. After 30 minutes, cells were collected by centrifugation at 300 *g*, 5 minutes, 4 °C and resuspended in 100 µL of Anti-human-kappa-light chain-FITC conjugated antibody (RRID: AB_259557, F-3761, Sigma-Aldrich), diluted at 1:100 in 2% BSA/PBS. Cells were incubated for 30 minutes on ice, and after a final centrifugation at 300 *g*, 5 minutes, 4 °C, they were resuspended with ice-cold PBS and analyzed with a Guava easyCyte flow cytometer instrument (Luminex).

Cell–cell interaction assay.. Targeted MDA-MB-468 and negative control cells (HEK293-6E) were stained with Vybrant™ DiD Cell-Labeling Solution and the CD89-expressing cells (HL-60) with Vybrant™ DiI Cell-Labeling Solution (both Fisher Scientific), for 20 min at 37 °C. 100,000 cells of each type were mixed and incubated in 100 µL of antibody dilution of graded concentrations, starting with 20 nM, in 2% BSA-PBS in 96-U-well plates for 30 min on ice. After centrifugation at 300 *g*, 5 min at 4 °C, cells were resuspended in 200 µL ice-cold PBS and analyzed with a Guava easyCyte flow cytometer (Luminex).

Internalization experiments.. Bispecific antibodies were labeled with Alexa Fluor 488 (A10235, Fisher Scientific) following manufacturer's instructions. Target cells were harvested and blocked as described above. 100,000 MDA-MB-468 cells/mL were incubated with 20 nM antibodies at 37°C for 4 h or overnight. After the incubation, the cells were detached at 300 *g* for 5 min at 4°C, rinsed once with 200 µL ice-cold PBS, and one parallel sample was incubated with 20 µg/mL anti-Alexa Fluor 488 antibody (RRID: AB_221544, A11094, Fisher Scientific), diluted in 2% BSA-PBS for 30 min on ice. After centrifugation, the cells were resuspended in 200 µL ice-cold PBS and analyzed with a Guava easyCyte flow cytometer (Luminex).

Neutrophil activation

Neutrophil isolation

Full blood was drawn into heparin tubes. 15 mL of blood were transferred to 50-ml-conical tube, mixed 10 times by repeated inversion with an equal volume of 3% Dextran and the mixture was allowed to set for 20 min at RT. Leukocyte-rich upper layer was aspirated with a sterile plastic pipette and transferred to a fresh conical tube. Leukocytes were pelleted by centrifugation at 500 *g* for 10 min at 4 °C with brake off, and the supernatant was discarded. Pellets were resuspended in 10 mL PBS and were pooled into a single 50-mL-conical tube to achieve a final volume of 35 mL, and the suspension was underlayed with 15 mL Ficoll-Hypaque. After centrifugation at 400 *g* for 30 min at RT, the polymorphonuclear cell-erythrocyte layer was isolated and resuspended in 10 mL sterile water for 28 s, an equal volume of 1.8% saline was added and cells were pelleted at 500 g for 5 min at RT. Finally, the cells were resuspended in 5 mL PBS, and their concentration was determined using a hemocytometer. Finally, neutrophils were resuspended in assay media (DMEM medium with 10% FCS and penicillin/streptomycin) at a final density of $8x10^6$/mL.

Neutrophil characterization

Isolated neutrophils were characterized using staining with anti-CD45, anti-CD16, and anti-CD66b antibodies (Stemcell 60018AZ, 60041FI.1, 60086FI.1). Cells were blocked at a density of $1x10^6$/mL with 10% FCS in PBS for 20 min at 4 °C. Aliquots of 100,000 cells were delivered into each well, centrifuged at 300 g, 5 min at 4°C, and incubated with 1 µL antibody, diluted in 100 µL 0.5% FCS in PBS for 20 min at 4°C. Cells were centrifuged and resuspended in 200 µL ice-cold PBS and 5000 events were measured with Guava easyCyte (Luminex). CD89 expression was measured as described above for HL-60 cells.

Cytotoxicity assay

Targeted MDA-MB-468 cells at a density of 10,000 cells/well were allowed to adhere overnight. Medium was removed and antibodies at 20 nM concentration, as well as the neutrophils at an effector-to-target ratio of 40:1 were added and incubated overnight in a humidified atmosphere at 37°C with 5% CO_2. Plates were then inverted and blotted dry, and 100 µL of complete culture medium containing CellTiter-Blue® Cell

Viability Assay Reagent (Promega), diluted 1:5, were added per well. Fluorescence at 560/590 nm was measured after 1.5-h incubation using Tecan Spark microplate reader (Tecan Instruments) after shaking the plate for 10 s. Quadruplicate samples were measured and, for HEK293-6E, used as control cells, triplicate samples were measured. Wells without any cells were considered background, and the cells without any antibody were considered not to experience any specific killing. Complete lysis was assessed in wells where 11 µL Lysis Solution (Promega) was added to the medium 45 min previously to the completion of the incubation. Additional controls, where no neutrophils were added, were performed to exclude the effect of direct toxicity of the compounds on the cells (in six parallels). Percent-specific killing of target cells was calculated according to the following equation: 100 − ((relative fluorescence units (RFU) of test wells-RFU of background wells) − (RFU of wells without antibody-RFU of background wells)/(RFU of wells without antibody-RFU of background wells)) and background were the wells with medium and substrate only.

Disclosure statement

VS, LP, SZ, LT and SB are current or past employees of Merck Healthcare KGaA.

Funding

Financial support by the company Merck Healthcare KGaA, Christian Doppler Society, Austrian Federal Ministry for Digital and Economic Affairs, and the National Foundation for Research, Technology, and Development is gratefully acknowledged (CD Laboratory for Innovative Immunotherapeutics, grant to GWK). This project was also supported by the Connective Base GmbH, BOKU Core Facility Biomolecular & Cellular Analysis, and BOKU Core Facility for Mass Spectrometry; Christian Doppler Forschungsgesellschaft [CD Laboratory for Innovative Immunotherapeutics].

ORCID

Veronica Natale http://orcid.org/0000-0003-0317-675X
Katharina Stadlbauer http://orcid.org/0000-0002-0724-5874
Florian Rüker http://orcid.org/0000-0001-7639-2686
Lukas Pekar http://orcid.org/0000-0001-9259-0965
Stefan Zielonka http://orcid.org/0000-0002-4649-2843
Stefan Becker http://orcid.org/0009-0008-0246-6804
Gordana Wozniak-Knopp http://orcid.org/0000-0002-4069-070X

List of abbreviations

ADCC	antibody-dependent cellular cytotoxicity
BLI	biolayer interferometry
BSA	bovine serum albumin
DSC	differential scanning calorimetry
EC_{50}	half maximal effective concentration
ECD	extracellular domain
EDTA	ethylenediaminetetraacetic acid
EGFR	epidermal growth factor receptor
ELISA	enzyme-linked immunosorbent assay
Fab	fragment antigen binding
FACS	fluorescence-activated cell sorting
Fc	fragment crystallizable
FcRn	neonatal Fc receptor
FCS	fetal calf serum
FcαRI	fragment crystallizable alpha receptor I
FUT8	fucosyltransferase 8
HEK	human embryonal kidney
Ig	immunoglobulin
NK cells	natural killer cells
PBS	phosphate-buffered saline
PEI	polyethyleneimine

RFU	relative fluorescence units
ROR1	receptor tyrosine kinase like orphan receptor 1
RT	room temperature
SEC	size exclusion chromatography
SEED	strand-exchange engineered domain
T_M	midpoint of transition temperature
TMB	3,3′,5,5′-3,3,5,5-tetramethylbenzidine

AI statement

The author did not use any artificial intelligence programs or systems during the preparation of this work. The authors take full responsibility for the publication's content.

Author contributions

VN, VS, SB, and GWK designed the study; VN, GH, KS, VS, and LP executed the experiments and analyzed the data; VN and GWK performed data visualization; FR, SZ, SB, LT, and GWK supervised the study; VN and GWK wrote the manuscript; all authors read and revised the manuscript.

Data availability statement

Data are available within the article or its supplementary materials.

References

1. Shuptrine CW, Surana R, Weiner LM. Monoclonal antibodies for the treatment of cancer. Semin Cancer Biol. 2012;22(1):3–13. doi: 10.1016/j.semcancer.2011.12.009.
2. Ferris RL, Jaffee EM, Ferrone S. Tumor antigen-targeted, monoclonal antibody-based immunotherapy: clinical response, cellular immunity, and immunoescape. J Clin Oncol. 2010;28(28):4390–4399. doi: 10.1200/JCO.2009.27.6360.
3. Wei J, Yang Y, Wang G, Liu M. Current landscape and future directions of bispecific antibodies in cancer immunotherapy. Front Immunol. 2022;13:1–21. doi: 10.3389/fimmu.2022.1035276.
4. Friedrich MJ, Neri P, Kehl N, Michel J, Steiger S, Kilian M, Leblay N, Maity R, Sankowski R, Lee H, et al. The pre-existing T cell landscape determines the response to bispecific T cell engagers in multiple myeloma patients. Cancer Cell. 2023;41(4):711–725.e6. doi: 10.1016/j.ccell.2023.02.008.
5. Beurskens FJ, Lindorfer MA, Farooqui M, Beum PV, Engelberts P, Mackus WJM, Parren P, Wiestner A, Taylor RP. Exhaustion of cytotoxic effector systems may limit monoclonal antibody-based immunotherapy in cancer patients. J Of Immunol. 2012;188(7):3532–3541. doi: 10.4049/jimmunol.1103693.
6. Shimabukuro-Vornhagen A, Gödel P, Subklewe M, Stemmler HJ, Schlößer HA, Schlaak M, Kochanek M, Böll B, von bergwelt-Baildon MS. Cytokine release syndrome. J Immunotherapy Cancer. 2018;6(1). doi: 10.1186/s40425-018-0343-9.
7. Karki R, Kanneganti TD. The 'cytokine storm': molecular mechanisms and therapeutic prospects. Trends In Immunol. 2021;42(8):681–705. doi: 10.1016/J.IT.2021.06.001.
8. Stockmeyer B, Dechant M, van Egmond M, Tutt AL, Sundarapandiyan K, Graziano RF, Repp R, Kalden JR, Gramatzki M, Glennie MJ, et al. Triggering FCα-receptor I (CD89) recruits neutrophils as effector cells for CD20-directed antibody therapy. The J Immunol. 2000;165(10):5954–5961. doi: 10.4049/JIMMUNOL.165.10.5954.
9. Van Egmond M, Bakema JE. Neutrophils as effector cells for antibody-based immunotherapy of cancer. Semin Cancer Biol. 2013;23(3):190–199. doi: 10.1016/J.SEMCANCER.2012.12.002.
10. Di Carlo E, Forni G, Lollini PL, Colombo MP, Modesti A, Musiani P. The intriguing role of polymorphonuclear neutrophils in antitumor reactions. Blood. 2001;97(2):339–345. doi:10.1182/blood.V97.2.339.
11. Gruijs M, Sewnath CAN, van Egmond M. Therapeutic exploitation of neutrophils to fight cancer. Semin Immunol. 2021;57:1–9. doi: 10.1016/j.smim.2021.101581.
12. Hao Y, Hu P, Zhang J. Genomic analysis of the prognostic effect of tumor-associated neutrophil-related genes across 15 solid cancer types: an immune perspective. Ann Transl Med. 2020;8(22):1507–1507. doi:10.21037/atm-20-6629.
13. Avtenyuk NU, Visser N, Bremer E, Wiersma VR. The neutrophil: the underdog that packs a punch in the fight against cancer. Int J Mol Sci. 2020;21(21):1–34. doi: 10.3390/ijms21217820.
14. Takai T. Roles of fc receptors in autoimmunity. Nat Rev Immunol. 2002;2(8):580–592. doi:10.1038/nri856.

15. Brandsma AM, Ten BT, Nederend M, Meulenbroek LAPM, Van Tetering G, Meyer S, Jansen JHM, Buitrago MAB, Nagelkerke SQ, Németh I, et al. Simultaneous targeting of FcγRs and FcαRI enhances tumor cell killing. Cancer Immunol Res. 2015;3(12):1316–1324. doi: 10.1158/2326-6066.CIR-15-0099-T.
16. Heemskerk N, Gruijs M, Robin Temming A, Heineke MH, Gout DY, Hellingman T, Tuk CW, Winter PJ, Lissenberg-Thunnissen S, Bentlage AEH, et al. Augmented antibody-based anticancer therapeutics boost neutrophil cytotoxicity. J Clin Invest. 2021;131(6):1–16. doi: 10.1172/JCI134680.
17. Borrok MJ, Luheshi NM, Beyaz N, Davies GC, Legg JW, Wu H, Dall'acqua WF, Tsui P. Enhancement of antibody-dependent cell-mediated cytotoxicity by endowing igG with FcαRI (CD89) binding. Mabs-austin. 2015;7(4):743–751. doi: 10.1080/19420862.2015.1047570.
18. Kelton W, Mehta N, Charab W, Lee J, Lee CH, Kojima T, Kang TH, Georgiou G. IgGA: a "cross-isotype" engineered human Fc antibody domain that displays both IgG-like and IgA-like effector functions. Chem Biol. 2014;21(12):1603–1609. doi: 10.1016/j.chembiol.2014.10.017.
19. Wines BD, Sardjono CT, Trist HM, Lay CS, Hogarth PM. The interaction of FcαRI with IgA and Its implications for ligand binding by immunoreceptors of the leukocyte receptor cluster. The J Immunol. 2001;166 (3):1781–1789. doi: 10.4049/jimmunol.166.3.1781.
20. Breedveld A, Van Egmond M. IgA and FcαRI: pathological roles and therapeutic opportunities. Front Immunol. 2019;10(MAR):1–20. doi: 10.3389/fimmu.2019.00553.
21. Mazor Y, Oganesyan V, Yang C, Hansen A, Wang J, Liu H, Sachsenmeier K, Carlson M, Gadre DV, Borrok MJ, et al. Improving target cell specificity using a novel monovalent bispecific IgG design. Mabs-austin. 2015;7 (2):377–389. doi: 10.1080/19420862.2015.1007816.
22. Fitzgerald J, Lugovskoy A. Rational engineering of antibody therapeutics targeting multiple oncogene pathways. Mabs-austin. 2011;3(3):299–309. doi:10.4161/MABS.3.3.15299.
23. Labrijn AF, Janmaat ML, Reichert JM, Parren PWHI. Bispecific antibodies: a mechanistic review of the pipeline. Nat Rev Drug Discov. 2019;18(8):585–608. doi: 10.1038/s41573-019-0028-1.
24. Davis JH, Aperlo C, Li Y, Kurosawa E, Lan Y, Lo KM, Huston JS. Seedbodies: fusion proteins based on strand-exchange engineered domain (SEED) CH3 heterodimers in an Fc analogue platform for asymmetric binders or immunofusions and bispecific antibodies. Protein Eng Des Sel. 2010;23(4):195–202. doi: 10.1093/protein/gzp094.
25. Sellmann C, Doerner A, Knuehl C, Rasche N, Sood V, Krah S, Rhiel L, Messemer A, Wesolowski J, Schuette M, et al. Balancing selectivity and efficacy of bispecific epidermal growth factor receptor (EGFR) x c-MET antibodies and antibody-drug conjugates. J Biol Chem. 2016;291(48):25106–25119. doi: 10.1074/jbc.M116.753491.
26. Kelton C, Wesolowski JS, Soloviev M, Schweickhardt R, Fischer D, Kurosawa E, McKenna SD, Gross AW. Anti-EGFR biparatopic-SEED antibody has enhanced combination-activity in a single molecule. Arch Biochem Biophys. 2012;526(2):219–225. doi: 10.1016/j.abb.2012.03.005.
27. Muda M, Gross AW, Dawson JP, He C, Kurosawa E, Schweickhardt R, Dugas M, Soloviev M, Bernhardt A, Fischer D, et al. Therapeutic assessment of SEED: a new engineered antibody platform designed to generate mono- and bispecific antibodies. Protein Eng Des Sel. 2011;24(5):447–454. doi: 10.1093/protein/gzq123.
28. Dietrich S, Gross AW, Becker S, Hock B, Stadlmayr G, Rüker F, Wozniak-Knopp G. Constant domain-exchanged fab enables specific light chain pairing in heterodimeric bispecific SEED-antibodies. Biochim Biophys Acta - Proteins Proteomics. 2020;1868(1):1–11. doi: 10.1016/j.bbapap.2019.07.003.
29. Krah S, Schröter C, Eller C, Rhiel L, Rasche N, Beck J, Sellmann C, Günther R, Toleikis L, Hock B, et al. Generation of human bispecific common light chain antibodies by combining animal immunization and yeast display. Protein Eng Des Sel. 2017;30(4):291–301. doi: 10.1093/protein/gzw077.
30. Bönisch M, Sellmann C, Maresch D, Halbig C, Becker S, Toleikis L, Hock B, Rüker F. Novel CH1:CLinterfaces that enhance correct light chain pairing in heterodimeric bispecific antibodies. Protein Eng Des Sel. 2017;30 (9):685–696. doi: 10.1093/protein/gzx044.
31. Heinkel F, Verstraete MM, Cao S, Li J, Farber P, Stangle E, Silva-Moreno B, Peng F, Dixit S, Boulanger MJ, et al. Engineering a pure and stable heterodimeric IgA for the development of multispecific therapeutics. Mabs-austin. 2022;14(1):1–13. doi: 10.1080/19420862.2022.2141637.
32. Kawamoto T, Sato JD, Le A, Polikoff J, Sato GH, Mendelsohn J. Growth stimulation of A431 cells by epidermal growth factor: identification of high-affinity receptors for epidermal growth factor by an anti-receptor monoclonal antibody. Proc Natl Acad Sci USA. 1983;80(5 I):1337–1341. doi: 10.1073/pnas.80.5.1337.
33. Rabinovich B, Dawson J, Zhou X, Becker S, Nanfack YF. WO 2023/019121 A3 - proteins that decouple T cell-mediated tumor cytotoxicity from release of pro-inflammatory cytokines. The Lens. [Accessed 2023 Sep 15]. https://www.lens.org/lens/patent/128-703-014-950-203/frontpage?l=en.
34. Palumbo C, Benvenuto M, Focaccetti C, Albonici L, Cifaldi L, Rufini A, Nardozi D, Angiolini V, Bei A, Masuelli L, et al. Recent findings on the impact of ErbB receptors status on prognosis and therapy of head and neck squamous cell carcinoma. Front Med. 2023;10:1–16. doi: 10.3389/fmed.2023.1066021.
35. Baskar S, Kwong KY, Hofer T, Levy JM, Kennedy MG, Lee E, Staudt LM, Wilson WH, Wiestner A, Rader C. Unique cell surface expression of receptor tyrosine kinase ROR1 in human B-cell chronic lymphocytic leukemia. Clin Cancer Res. 2008;14(2):396–404. doi: 10.1158/1078-0432.CCR-07-1823.

36. Zhang S, Chen L, Cui B, Chuang HY, Yu J, Wang-Rodriguez J, Tang L, Chen G, Basak GW, Kipps TJ, et al. ROR1 Is expressed in human breast cancer and associated with enhanced tumor-cell growth. PLOS One. 2012;7 (3):31127. doi: 10.1371/JOURNAL.PONE.0031127.
37. Yamaguchi T, Yanagisawa K, Sugiyama R, Hosono Y, Shimada Y, Arima C, Kato S, Tomida S, Suzuki M, Osada H, et al. NKX2-1/TITF1/TTF-1-Induced ROR1 Is required to sustain EGFR survival signaling in lung adenocarcinoma. Cancer Cell. 2012;21(3):348–361. doi: 10.1016/j.ccr.2012.02.008.
38. Herr AB, Ballister ER, Bjorkman PJ. Insights into IgA-mediated immune responses from the crystal structures of human FcαRI and its complex with IgA1-Fc. Nature. 2003;423(6940):614–620. doi: 10.1038/nature01685.
39. Burmeister WP, Gastinel LN, Simister NE, Blum ML, Bjorkman PJ. Crystal structure at 2.2 Å resolution of the MHC-related neonatal Fc receptor. Nature. 1994;372(6504):336–343. doi: 10.1038/372336a0.
40. Boehm MK, Woof JM, Kerr MA, Perkins SJ. The fab and fc fragments of IgA1 exhibit a different arrangement from that in IgG: a study by X-ray and neutron solution scattering and homology modelling 1 1 edited by R. Huber. J Mol Biol. 1999;286(5):1421–1447. doi: 10.1006/jmbi.1998.2556.
41. Waterhouse A, Bertoni M, Bienert S, Studer G, Tauriello G, Gumienny R, Heer FT, De Beer TAP, Rempfer C, Bordoli L, et al. SWISS-MODEL: homology modelling of protein structures and complexes. Nucleic Acids Res. 2018;46(W1):W296–W303. doi: 10.1093/NAR/GKY427.
42. Posgai MT, Tonddast-Navaei S, Jayasinghe M, Ibrahim GM, Stan G, Herr AB. FcαRI binding at the IgA1 CH2–CH3 interface induces long-range conformational changes that are transmitted to the hinge region. Proc Natl Acad Sci USA. 2018;115(38):E8882–E8891. doi: 10.1073/pnas.1807478115.
43. Putnam FW, Liu YSV, Low TLK. Primary structure of a human IgA1 immunoglobulin. IV. Streptococcal IgA1 protease digestion, fab and Fc fragments, and the complete amino acid sequence of the α1 heavy chain. J Biol Chem. 1979;254(8):2865–2874. doi: 10.1016/s0021-9258(17)30153-9.
44. Traxlmayr MW, Hasenhindl C, Hackl M, Stadlmayr G, Rybka JD, Borth N, Grillari J, Rüker F, Obinger C. Construction of a stability landscape of the CH3 domain of human IgG1 by combining directed evolution with high throughput sequencing. J Mol Biol. 2012;423(3):397–412. doi: 10.1016/j.jmb.2012.07.017.
45. Brunke C, Lohse S, Derer S, Peipp M, Boross P, Kellner C, Beyer T, Dechant M, Royle L, Liew LP, et al. Effect of a tail piece cysteine deletion on biochemical and functional properties of an epidermal growth factor receptor-directed IgA2 min(1) antibody. Mabs-austin. 2013;5(6):936–945. doi: 10.4161/mabs.26396.
46. Lohse S, Meyer S, Meulenbroek LAPM, Jansen JHM, Nederend M, Kretschmer A, Klausz K, Möginger U, Derer S, Rösner T, et al. An anti-EGFR IgA that displays improved pharmacokinetics and myeloid effector cell engagement in vivo. Cancer Res. 2016;76(2):403–417. doi: 10.1158/0008-5472.CAN-15-1232.
47. Lohse S, Brunke C, Derer S, Peipp M, Boross P, Kellner C, Beyer T, Dechant M, Van Der Winkel JGJ, Leusen JHW, et al. Characterization of a mutated IgA2 antibody of the m(1) allotype against the epidermal growth factor receptor for the recruitment of monocytes and macrophages. J Biol Chem. 2012;287 (30):25139–25150. doi: 10.1074/jbc.M112.353060.
48. Varki A, Cummings RD, Aebi M, Packer NH, Seeberger PH, Esko JD, Stanley P, Hart G, Darvill A, Kinoshita T, et al. Symbol nomenclature for graphical representations of glycans. Glycobiology. 2015;25(12):1323–1324. doi: 10.1093/GLYCOB/CWV091.
49. Mehta AY, Cummings RD, Wren J. GlycoGlyph: a glycan visualizing, drawing and naming application. Bioinformatics. 2020;36(11):3613–3614. doi: 10.1093/bioinformatics/btaa190.
50. Brandsma AM, Bondza S, Evers M, Koutstaal R, Nederend M, Marco Jansen JH, Rösner T, Valerius T, Leusen JHW, Ten Broeke T. Potent Fc receptor signaling by IgA leads to superior killing of cancer cells by neutrophils compared to IgG. Front Immunol. 2019;10(APR):1–14. doi: 10.3389/fimmu.2019.00704.
51. Oortwijn BD, Roos A, van der Boog PJM, Klar-Mohamad N, van Remoortere A, Deelder AM, Daha MR, van Kooten C. Monomeric and polymeric IgA show a similar association with the myeloid FcαRI/CD89. Mol Immunol. 2007;44(5):966–973. doi: 10.1016/J.MOLIMM.2006.03.014.
52. Bakema JE, Van Egmond M. The human immunoglobulin A Fc receptor FcαRI: a multifaceted regulator of mucosal immunity. Mucosal Immunol. 2011;4(6):612–624. doi: 10.1038/mi.2011.36.
53. Senior BW, Woof JM. Effect of mutations in the human immunoglobulin A1 (IgA1) hinge on its susceptibility to cleavage by diverse bacterial IgA1 proteases. Infect Immun. 2005;73(3):1515–1522. doi:10.1128/IAI.73.3.1515-1522.2005.
54. Woof JM, Russell MW. Structure and function relationships in IgA. Mucosal Immunol. 2011;4(6):590–597. doi: 10.1038/mi.2011.39.
55. Heineke MH, van Egmond M. Immunoglobulin A: magic bullet or trojan horse?. Eur J Clin Invest. 2017;47 (2):184–192. doi: 10.1111/eci.12716.
56. Stip MC, Jansen JHM, Nederend M, Tsioumpekou M, Evers M, Olofsen PA, Meyer-Wentrup F, Leusen JHW. Characterization of human Fc alpha receptor transgenic mice: comparison of CD89 expression and antibody-dependent tumor killing between mouse strains. Cancer Immunol, Immunother : CII. 2023;72 (9):3063–3077. doi: 10.1007/s00262-023-03478-4.
57. Xu L, Yang B, Zhang J, Liu S, Zhang Q, Ran L, Li B. Targeting ALPPL2 with a novel CD89 bispecific antibody reprograms macrophages to enhance anti-tumor immunity. Biochem Biophys Res Commun. 2025;762:762. doi: 10.1016/J.BBRC.2025.151761.

58. Treffers LW, ten Broeke T, Rosner T, Marco Jansen JH, van Houdt M, Kahle S, Schornagel K, Verkuijlen PJJH, Prins JM, Franke K, et al. IgA-mediated killing of tumor cells by neutrophils is enhanced by CD47–SIRPA checkpoint inhibition. Cancer Immunol Res. 2020;8(1):120–130. doi: 10.1158/2326-6066.CIR-19-0144.
59. Hellwig SMM, Van Spriel AB, Schellekens JFP, Mooi FR, Van de Winkel JGJ, Clements JD. Immunoglobulin A-mediated protection against bordetella pertussis infection. Infect Immun. 2001;69(8):4846–4850. doi: 10.1128/IAI.69.8.4846-4850.2001.
60. Balu S, Reljic R, Lewis MJ, Pleass RJ, McIntosh R, van Kooten C, van Egmond M, Challacombe S, Woof JM, Ivanyi J. A novel human IgA monoclonal antibody protects against tuberculosis. The J Immunol. 2011;186(5):3113–3119. doi: 10.4049/jimmunol.1003189.
61. Gong S, Lakhashe SK, Hariraju D, Scinto H, Lanzavecchia A, Cameroni E, Corti D, Ratcliffe SJ, Rogers KA, Xiao P, et al. Cooperation between systemic IgG1 and mucosal dimeric IgA2 monoclonal anti-HIV env antibodies: passive immunization protects indian rhesus macaques against mucosal SHIV challenges. Front Immunol. 2021;12:1–10. doi: 10.3389/fimmu.2021.705592.
62. Nielsen TB, Yan J, Slarve M, Li R, Junge JA, Luna BM, Wilkinson I, Yerramalla U, Spellberg B. Development of a bispecific antibody targeting clinical isolates of acinetobacter baumannii. The J Infect Dis. 2023;227(9):1042–1049. doi: 10.1093/infdis/jiac499.
63. Pyzik M, Kozicky LK, Gandhi AK, Blumberg RS. The therapeutic age of the neonatal Fc receptor. Nat Rev Immunol. 2023;23(7):415–432. doi: 10.1038/s41577-022-00821-1.
64. Mester S, Evers M, Meyer S, Nilsen J, Greiff V, Sandlie I, Leusen J, Andersen JT. Extended plasma half-life of albumin-binding domain fused human IgA upon pH-dependent albumin engagement of human FcRn in vitro and in vivo. Mabs-austin. 2021;13(1):1–13.e1893888. doi: 10.1080/19420862.2021.1893888.
65. Mester S, Chan C, Lustig M, Foss S, Jansen JHM, Leangen Herigstad M, Evers M, Nilsen J, Reiding KR, Damen JMA, et al. Engineering of anticancer human immunoglobulin A equipped with albumin for enhanced plasma half-life. PNAS Nexus. 2025;4(2). doi: 10.1093/PNASNEXUS/PGAF042.
66. García-García A, Serna S, Yang Z, Delso I, Taleb V, Hicks T, Artschwager R, Vakhrushev SY, Clausen H, Angulo J, et al. FUT8-Directed core fucosylation of N-glycans is regulated by the glycan structure and protein environment. ACS Catal. 2021;11(15):9052–9065. doi: 10.1021/acscatal.1c01698.
67. Gomes MM, Wall SB, Takahashi K, Novak J, Renfrow MB, Herr AB. Analysis of IgA1 N-glycosylation and its contribution to FcαRI binding. Biochemistry. 2008;47(43):11285–11299. doi: 10.1021/bi801185b.
68. Amore A, Cirina P, Conti G, Brusa P, Peruzzi L, Coppo R. Glycosylation of circulating IgA in patients with IgA nephropathy modulates proliferation and apoptosis of mesangial cells. J Am Soc of Nephrol : JASN. 2001;12(9):1862–1871. doi: 10.1681/asn.v1291862.
69. Stip MC, Evers M, Nederend M, Chan C, Reiding KR, Damen MJ, Heck AJR, Koustoulidou S, Ramakers R, Krijger GC, et al. IgA antibody immunotherapy targeting GD2 is effective in preclinical neuroblastoma models. J for Immunother of Cancer. 2023;11(7):e006948. doi: 10.1136/JITC-2023-006948.
70. Crescioli S, Jatiani S, Moise L. With great power, comes great responsibility: the importance of broadly measuring Fc-mediated effector function early in the antibody development process. Mabs-austin. 2025;17(1). doi: 10.1080/19420862.2025.2453515.